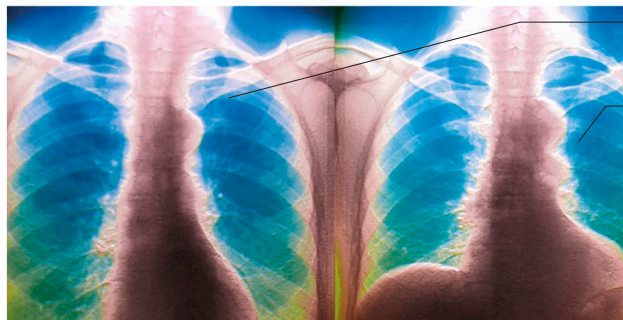




Chest cavity expands as diaphragm and intercostal muscles contract

Chest cavity decreases in size as diaphragm and intercostal muscles relax

DIAPHRAGM MOVEMENT

The abdominal contents (dark area at the bottom of this X-ray) are flattened by the diaphragm muscle during inhalation (left) and then rise up during exhalation (right).

EXHALATION

Breathing out is largely passive. Like a stretched elastic band, the enlarged lungs recoil and shrink when the diaphragm, intercostals, and other inhalation muscles relax, and abdominal pressure pushes the diaphragm upward. Forced expiration brings further muscles into play, to actively compress the lungs beyond their usual resting volume.

Lung
Shrinks with diminishing chest cavity volume

Diaphragm
Relaxes and pushes up into dome shape

Trachea
Stiff cartilage rings keep airways open under negative pressure

Sternum
Moves down and inward as ribs return to resting position

Internal intercostals
Pull ribs down for forced exhalation

Ribs
Are drawn in and tilt down

Rectus abdominis
Pulls on ribs five to seven and sternum, to depress ribcage and assist expiration

